# AES (Advanced Enviro-Septic$^{TM}$) Design and Installation Manual


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"Always The First Option"
ADVANCED ENVIRO-SEPTIC™

**Technical support**
Environment Technology provides technical support to all individuals using Presby Environmental products. For questions about products or the information in this manual, please phone 03 9707 979, visit the website www.et.nz or email info@et.nz

# Section A
# Introduction

**Background**

AES (Advanced Enviro-Septic™) is a passive Advanced Secondary Wastewater Treatment system tested as producing treated effluent of BOD(mg/L) ≤10 and TSS (mg/l) ≤10

Liquid that exits from a conventional septic tank (effluent) contains suspended solids that can cause other types of septic systems to fail prematurely. Solids can overload bacteria, cut off aeration required for bacterial activity, and/or seal the underlying soil.

**Our unique system components**

The AES system is a product consisting of three components installed in a sand bed following a standard a septic tank: -

1. A corrugated, perforated, high-density plastic pipe with a unique series of ridges on the peak of each corrugation and plastic "skimmers" extending into the pipe's interior.
2. A thick mat of randomly oriented plastic fibres surrounding the pipe.
3. A special non-woven geo-textile plastic fabric around the mat of fibres.

**What our system does**

By utilizing simple, yet effective, natural processes the AES system treats septic tank effluent in a manner that prevents solids from entering surrounding soils, increases system aeration, and provides a greater bacterial area (mat) than traditional systems.

**Why our system excels**

- Requires less fill
- Installs more easily and quickly than traditional systems
- Eliminates the need for expensive washed stone
- Adapts easily to both commercial and residential sites
- Uses a protected receiving surface
- Increases system performance and longevity
- Tests environmentally safer than conventional systems
- Recharges groundwater more safely than conventional systems

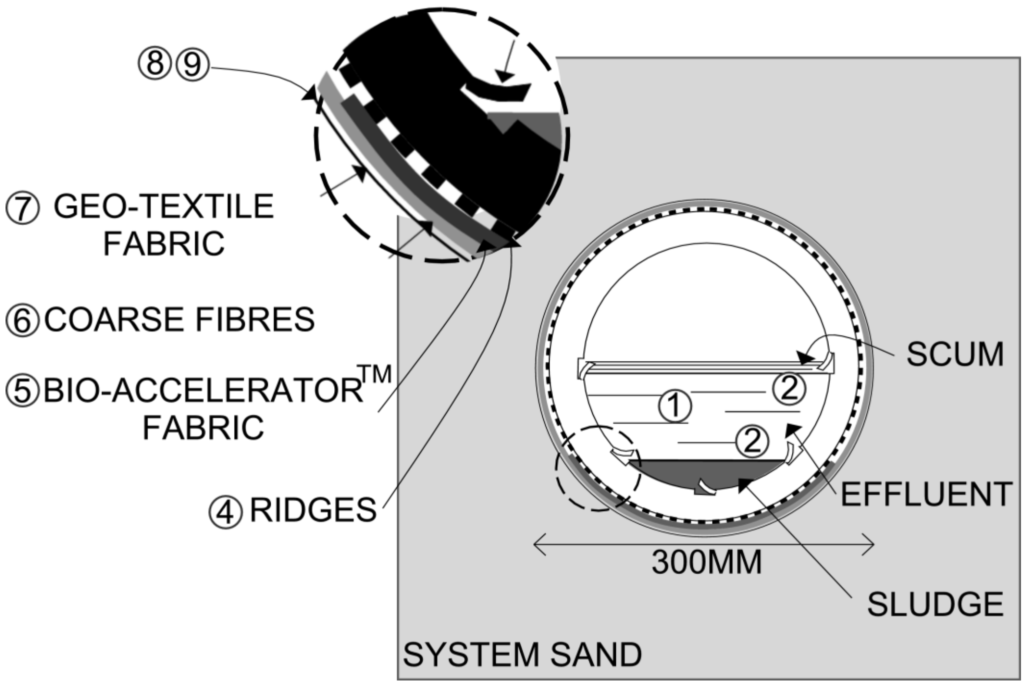
**System advantages**

- The AES system retains solids in its pipe and provides multiple bacterial surfaces to treat effluent prior to its contact with the soil.
- The continual cycling of effluent (the rising and falling of liquid inside the pipe) enhances bacterial growth.
- No other leaching system design offers this functionality.
- Our systems excel because they are more energy efficient, last longer, and have a minimal environmental impact.
- Costs less than traditional installation products and materials
- No on-going energy, servicing, mechanical or electronic repair costs
- Requires a smaller area
- Eliminates "septic mounds" through sloping system installations
- Adapts to difficult sites

# Section B
# Definitions of Terms

| | |
|---|---|
| **Centre-to-centre Spacing** | Centre-to-centre spacing is the horizontal distance from the centre of one line to the centre of the adjacent line. |
| **Coupling** | A Coupling is a fitting that joins two pieces of Advanced Enviro-Septic™ pipe. |
| **Design Flow** | Design Flow is the determined LPD flow as dictated by a recognised Standard or other publication. |
| **Differential venting** | Differential venting is a method of venting an Advanced Enviro-Septic™ system utilizing high and low vents (Oxygen Demand Vents). |
| **Advanced Enviro-Septic™ pipe** | An AES pipe is a single unit of pipe, 3m in length with an outside diameter of 300mm and a storage capacity of approximately 220 litres. |
| **High and low Vents** | High and Low (ODV) vents are pipes used in differential venting. Often the high vent is at the house - minimum 80mm diameter - and the low vent – minimum 100mm diameter - is on the last line of pipes in a Serial Distribution system. |
| **Level system** | A Level System is a system in which rows of AES pipes are installed at the same elevation. |
| **LPD** | LPD is an abbreviation for Litres Per Day. |
| **Offset adapter** | An Offset Adapter is an end cap fitted with a 100mm hole offset at the 12 o'clock position. |
| **Raised connection** | A Raised Connection is a 100mm PVC pipe arrangement used to connect lines of Advanced Enviro-Septic™ pipe to maintain the correct liquid level inside each line. |
| **Row** | A row is a number of sections of AES pipes in one linbeal direction, connected by couplings with an offset adapter on the inlet end and an offset adapter or end cap on the opposite end. |
| **Section** | A section is the length of pipe between two off-set adapters |
| **Serial Distribution** | Serial Distribution is a group of AES rows connected with a raised and/or drop connection. |
| **Sub-System** | A Sub-System is when a distribution box is used to split the system into 2 or more 'sub-systems' |
| **System Sand** | System Sand is a gravelly, coarse sand that meets the particle size requirements described in Section E below |

# Section C
# Advanced Enviro-septic$^{TM}$ Treatment Process




NOTE: System Sand is installed to a depth of 300mm below the AES pipes and 150mm above and between the AES pipes. In less permeable soils the lower 150mm of the sand bed is extended to provide the required infiltration area.

| | |
|---|---|
| Stage 1: | Warm effluent enters the pipe and is cooled to ground temperature |
| Stage 2: | Suspended solids separate from the cooled liquid |
| Stage 3: | Skimmers further capture the grease and suspended solids from the exiting effluent |
| Stage 4: | Pipe ridges allow the effluent to flow without interruption around the circumference of the pipe and aid in cooling |
| Stage 5: | Bio-Accelerator$^{TM}$ fabrics screen additional solids from the effluent and accelerates Biomat development |
| Stage 6: | A mat of coarse random fibres separates more suspended solids from the effluent |
| Stage 7: | Effluent passes into the geo-textile fabrics and grows a protected bacterial surface |
| Stage 8: | Sand wicks liquid from the geo-textile fabrics and enables air to transfer to the bacterial surface |
| Stage 9: | The fabrics and fibres provide a large bacterial surface to break down solids |
| Stage 10: | An ample air supply and fluctuating liquid levels increase bacterial efficiency |

# Section D
# Installation, Handling, and Storage Guidelines

**Introduction**

This page contains guidelines that must be observed while installing, handling, and storing Advanced Enviro-Septic$^{TM}$ products.

**Site Preparation**

Here are some site preparation guidelines.

- Remove topsoil, roots, and organic matter under the required sand area of a proposed system, including the slope extensions of raised systems.
- Maintain the existing characteristics of the underlying soil as much as possible. Avoid machine compaction.
- Add the sand fill on the same day that the leach area is excavated.
- Do not allow water to run into or over the system during construction.
- Do not work wet or frozen soils.
- Do not smear or compact soils while preparing site.
  Note: It is not necessary for the leach area to be smooth when the site is prepared.

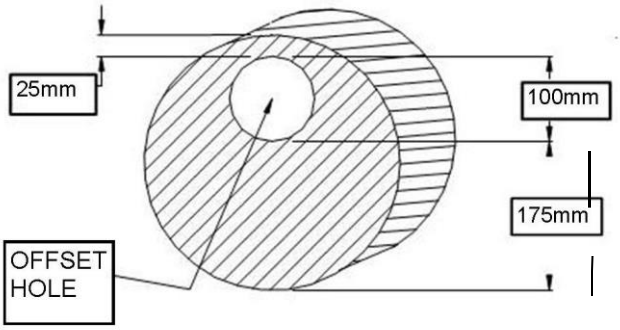
**System components**

Here is a picture of the Advanced Enviro-Septic$^{TM}$ components.

Note: Keep mud, grease, oil etc from all system components. Avoid dragging pipe through wet or muddy areas.

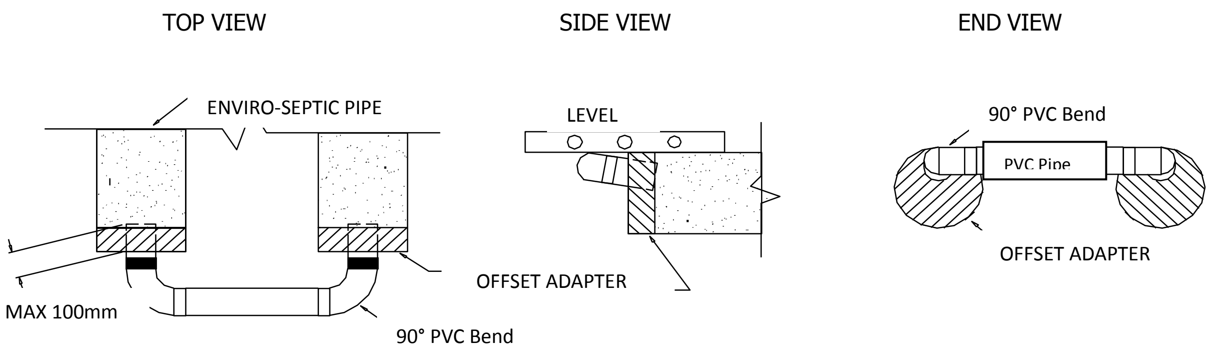
## Installation, Handling, and Storage Guidelines… continued

Use raised connections: Raised connections consist of offset adapters, 100mm PVC pipe and pipe elbows. They enable greater liquid storage capacity and increase the bacterial surfaces being developed. Use raised connections to connect lines of Enviro-Septic$^{TM}$ pipe. Here is a diagram along with some installation notes:



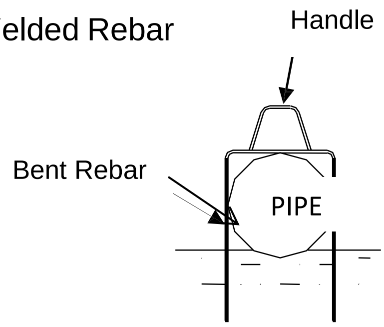

**Installation Notes:**

1. **Insert PVC pipe no more than 100mm into the offset adapter to prevent air locking.**
2. **Install the raised connection so that the top of the 90° bend is level with the top of the offset adapter.**
3. **Pack sand under and around the raised connection to prevent movement.**

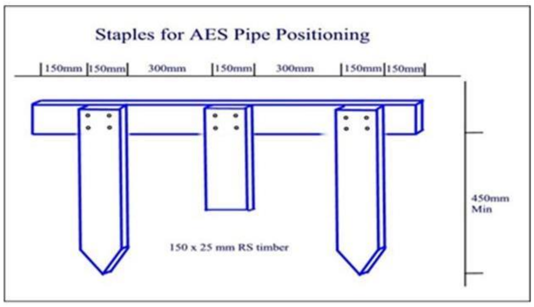



**Line spacers** While sand may be used to keep pipe in place while covering, simple tools may also be constructed for this purpose. Here are two examples. One is made from rebar, the other from wood.

Welded Rebar




Wooden frame




<u>Caution</u>: Remove all tools used as line spacers before final covering

# Installation, Handling, and Storage Guidelines… continued

**Soil compaction**

Minimize machine movement to avoid soil compaction and destruction of the soil structure under and around the system. Be especially careful not to compact soil on the down slope side of the system.

**Backfilling and final grading**

Spread a minimum of 150mm of system sand over the pipe. Spread the remaining fill. Final grading should shed water away from the system.

Note: A tracked vehicle may be used to spread the system sand and topsoil as long as it maintains at least 300mm of cover over the pipe.

**Erosion control**

Protect the site from erosion by proper grading, mulching, seeding, and control of runoff.

**Storage**

The outer fabric of the Advanced Enviro-Septic™ pipe is ultra-violet stabilized. However, the protection breaks down after a period of time in direct sunlight. To prevent damage to the fabric, cover the pipe with an opaque tarp.

Store pipe on high and dry areas to prevent surface water and soil from entering the pipes or contaminating the fabric prior to installation.

# Section E
# Sand and Fill Requirements

**Introduction**

This page describes the sand and fill requirements for the Advanced Enviro-Septic™ System.

**System sand**

All configurations of Advanced Enviro-Septic® require a minimum of 150mm of system sand surrounding the circumference of the pipe. This sand, typically gravelly coarse sand, must adhere to the following percentage and quality restrictions.

Percentage Restrictions
35% or less of the total sand may be gravel.
40%-90% of the total sand is to be coarse and very coarse sand.
Gravel Quality Restrictions
No gravel is to exceed 19mm in diameter.
No gravel is smaller than 2mm in diameter.
Coarse Sand Quality Restrictions
No coarse sand is smaller than 0.5mm in diameter.

ASTM Standard: C-33 (concrete sand) meets the above requirements. In addition the fine sand restrictions outlined below are necessary for ASTM C-33 sand to be used as System Sand

Fine Sand Quality Restrictions
No more than 2% of the total sand may pass through a 75 μm sieve.

**Sand fill and clean fill**

When AES beds are installed in natural sand this sand can be used as sand fill in the lower 150mm of the AES bed, provided it meets the fine sand quality restrictions above. Note: System sand may also be used as sand fill.

**Clean fill is the material used to complete the system.**

**Raised system fill extensions**

Raised systems require fill extensions with 3:1 (horizontal:vertical) batters.

**Sloping AES beds**

AES beds sloping greater than 5% require the system sand area to extend a minimum of 1m beyond the lowest line of Advanced Enviro-Septic™ pipes on the down-slope side.

See also 'AES System Sand' information sheet available from Environment Technology Ltd.

To find out if sand is suitable as 'system sand' send 2 cups of sand to our office and we can run a simple test on it.

Send sand to: Environment Technology
14 Onekaka Iron Works Rd
Takaka 7182

# Section F
# Single Level System Configurations

Advanced Enviro-Septic™ systems may be designed in a variety of unusual shapes such as curved, trapezoidal, or L-shaped to provide optimum design flexibility to address the challenges of each site.

**Sloping systems**

The percentage of slope refers to the slope of the Advanced Enviro-Septic™ system, not the existing terrain. The slope of the system and the existing terrain are not required to be equal. A sloping system can be designed with more than one distinct slope and/or center-to-center pipe spacing in the same system.

**Line orientation**

Advanced Enviro-Septic™ lines must be laid level and should run parallel to contours (perpendicular to sloping terrain) if possible.

**Velocity reduction**

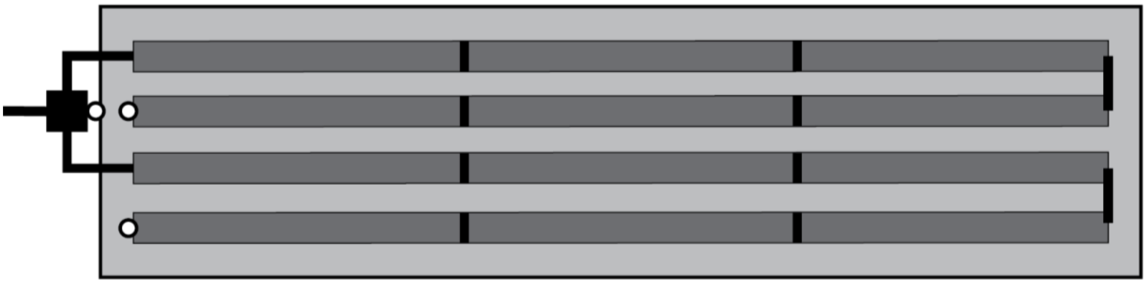
If piping from the septic tank to Advanced Enviro-Septic™ is excessively steep, or if effluent is pumped to the AES pipes a velocity reducer before the AES pipe inlet is necessary to reduce turbulence inside the AES pipe. A distribution box with a baffle or a100mm pipe with 90' bend may be an adequate velocity reducer.

## Basic Serial System – Level in-ground

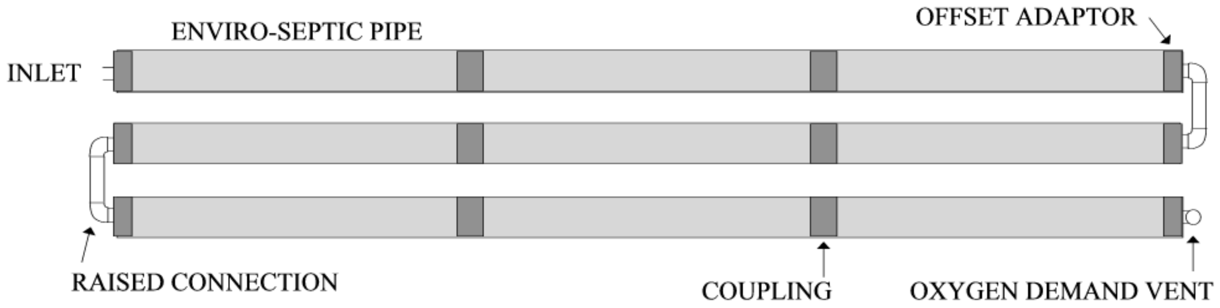
A basic serial system is a series of lines of Advanced Enviro-Septic$^{TM}$ connected by raised connections.

The following diagram shows One System comprising 2 sections, forming 2 rows

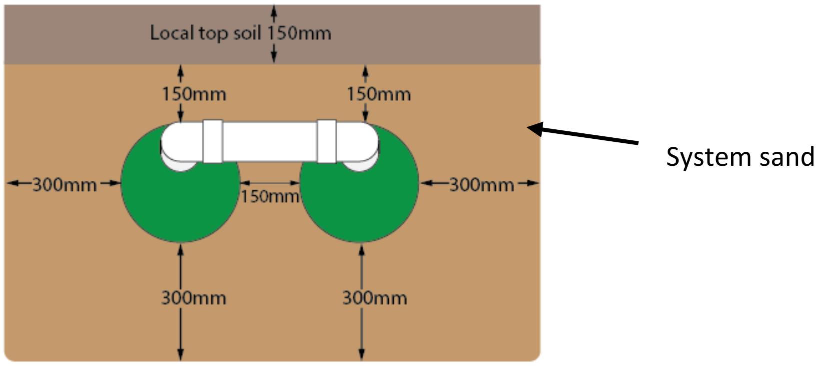
The following diagram shows Two Subsystems of 2 Sections forming 4 Rows in total



**Plan**


ENVIRO-SEPTIC PIPE
OFFSET ADAPTOR
INLET
RAISED CONNECTION
COUPLING
OXYGEN DEMAND VENT

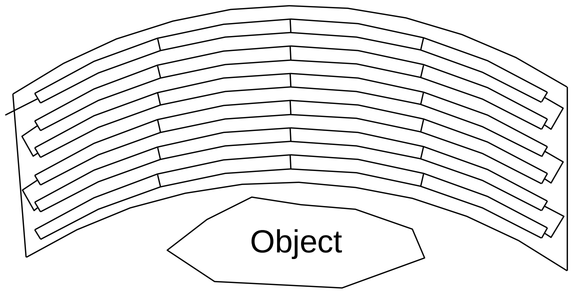
**Elevation**


Local top soil 150mm
150mm
150mm
300mm
150mm
300mm
300mm
300mm
System sand

# Non-Conventional Configurations

**Curves**

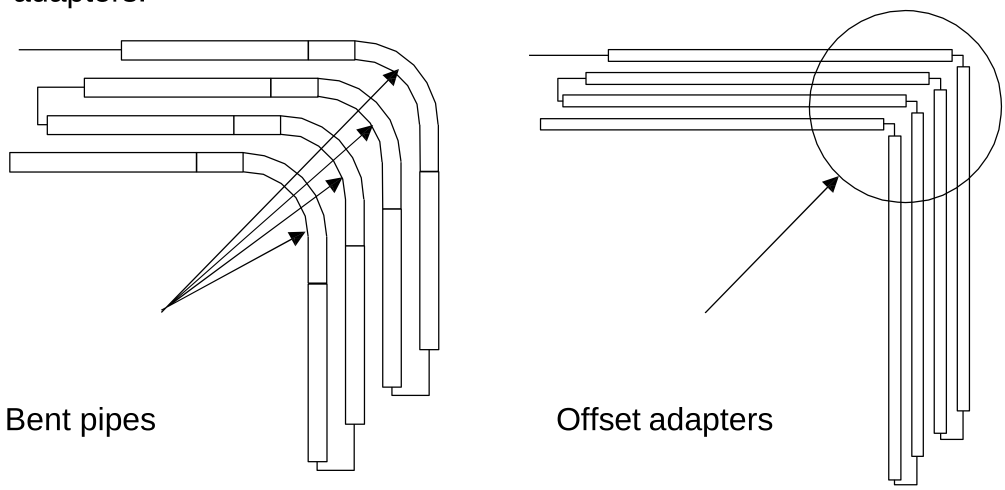
Curved configurations work well around objects, setbacks, and slopes.




Note: Multiple curves can also be used.

**Angles**

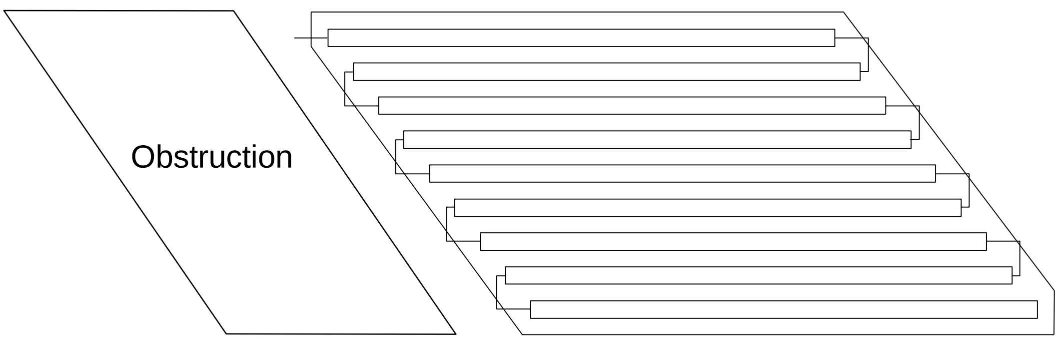
Angled configurations generally have one or more specific bends. Lines are angled by bending pipes or through the use of offset adapters.




Note: A 3m length of pipe may take a 90° bend.

**Trapezoids**

This system is trapezoidal to fit a particular slope or terrain feature.



# Section G
# System Rejuvenation and Expansion

**Introduction**

This section covers procedures for rejuvenating failing systems and explains how to expand existing systems.

**Definition: failing system**

System failures, almost without exception, are related to the conversion of bacteria from an aerobic to an anaerobic state. Flooding, improper venting, alteration or improper depth of soil, sudden use changes, introduction of chemicals or medicines, and a variety of other conditions can contribute to this.

**Rejuvenating failing systems**

Failing systems need to be returned from an anaerobic to an aerobic state. Most systems can be put back on line and not require costly removal and replacement by using the following procedure.

1. Determine the problem causing system failure and repair.
2. Excavate one end of all the lines and remove the end cap or offset adapter.
3. Drain the lines.
4. If foreign matter has entered the system, flush the pipes.
5. Safeguard the open excavation.
6. Guarantee a passage of air through the system.
7. Allow all lines to dry for a minimum of 72 hours.
8. Re-assemble the system to its original design configuration.

**System expansion**

Advanced Enviro-Septic™ systems are easily expanded by adding equal lengths of pipe to each line of the original design or by adding additional equal sections.

**Re-usable pipe**

Advanced Enviro-Septic™ components are not biodegradable and may be reused. In cases of improper installation it may be possible to excavate, clean, and reinstall all system components.

**System replacement**

If system components require replacement, simply remove the existing pipe and the contaminated sand and replace with new pipe and sand.

Your suggestions and comments are welcome. Please contact us at
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